ing thermal efficiency by 5-10 per cent. In her budget this year, finance minister Nirmala Sitharaman announced that National Thermal Power Corporation (NTPC) and Bharat Heavy Electricals Ltd (BHEL) will set up an 800 MW AUSC thermal power plant with indigenous technology jointly developed by NTPC, BHEL and the Indira Gandhi Centre for Atomic Research. In the private sector, the Adani Group is also planning massive investments in AUSCs.

HELLO SUNSHINE

One of the best alternatives for India is to go solar—in a big way. And this is one area where the Modi government has done creditable work. The installed capacity of solar energy, at 82.64 GW (as on April 30), is at 19 per cent of the total, the highest green source of energy in the country. The target is to set up 450 GW capacity by 2030. The challenge is the over-reliance on imports for critical raw material as well as machinery for production. Manufacturers import up to 70 per cent of the raw material and almost all the machinery for the production of photovoltaic panels from China. Solar power is also space-intensive, which is difficult in a country where land is premium. Experts like Sarabhai advocate that India seriously explore decentralising power generation to reduce costs of storage and transportation while scaling up capabilities. Reliance Industries recently announced plans to commission its first solar giga factory this year. To popularise solar energy adoption, the government had announced the PM Surya Ghar Muft Bijli Yojana in February 2024, with the ambitious aim of installing one million solar rooftops, incentivising individuals with free electricity of up to 300 units every month.

Biofuels like ethanol from agricultural waste can serve as another crucial resource in the country's search for self-sustainability, especially in rural areas. "India is naturally endowed with a lot of raw material, expertise and ability to lead in this area," says Rita Roychowdhary, Chief Executive—Climate Change & Sustainability Services, ECube Investment Advisors in Mumbai. "The capability and demand for this will have to be created at the local and state levels. The challenge is the raw material, which is segregated all across the country. There are entrepreneurship opportunities across the value chain." If the government incentivises the end use, she adds, this sector will get a boost as private companies enter the space and existing local players grow. India is also a part of the Global Biofuel Alliance, which allows it access to the latest technology. Biomass is also a source of hydrogen, which can also be channelised to generate green energy.

TAKEAWAYS

- Increase efficiency of thermal power plants and reduce emission in coal mining, storage and transport through use of tech
- Decentralise solar power generation to cut storage/transport costs
- Stabilise grid for hydrogen power production

PROJECT GREEN HYDROGEN

Sourced from fossil fuels, biomass and a mix of water and renewable energy, hydrogen has emerged as the most promising green fuel of the future. Especially Green Hydrogen, which is produced with renewable energy. Currently, natural gas is the primary source for the production of hydrogen, contributing approximately 75 per cent to the global hydrogen output of around 70 million tonnes annually. Abundant renewable energy resources, a stable grid infrastructure, extensive coastline and ports, strong engineering and construction capabilities, and supportive legislation place India in a favourable position to export green hydrogen to the world. Private sector giants have announced massive investments in green hydrogen plants. So, Adani Green has announced an investment of Rs 4.15 lakh crore in constructing green hydrogen production facilities, advancing technology, and establishing essential infrastructure. Similarly, Reliance Industries Ltd has pledged around Rs 80,000 crore for the same.

India's Green Hydrogen Mission, launched in 2022, has set a target of 5 MT per annum (one tonne of hydrogen delivers about 33 MWh and a million tonnes about 33 terawatt hours). For this, India will need an additional capacity of 125 GW of renewable energy to produce green hydrogen. A May 2024 report titled 'India's Green Hydrogen Revolution' by the Ministry of New and Renewable Energy along with EY states that "India has Pumped Hydro Storage (PHS), which acts like a large-scale 'water battery', storing excess energy from renewable sources (like solar and wind) during off-peak hours and releasing it back during high demand, contributing towards round-the-clock availability of renewable energy for Green Hydrogen production". The National Electricity Plan 2022 estimates the need for 18.8 GW of PHS by 2032. Along with PHS, a Battery Energy Storage System (BESS) is being developed on a large scale to enable the availability of renewable energy.

Yet, challenges abound here too. Electrolysers, the core equipment for green hydrogen production, require a stable power supply to operate efficiently and prevent damage. The problem is that since green hydrogen is produced primarily from renewables, fluctuations in generation can disrupt production, which requires an effective management of the grid. The other hurdles are the high cost of production, storage and transportation, as also the high requirement of water for electrolysis, which is the most advanced technology for the production of hydrogen.

India has the resolve, and the resources, for greening its future. All it needs is to step up the implementation. ■